

Maxstar® 400 and 800

**TIG/Stick Welding
Power Source**



Quick Specs



Industrial Applications

Precision metal fabrication
Tube mills
Pipe and tube fabrication
Tool and die
Exotic material fabrication
Pressure vessel fabrication

Processes

TIG (GTAW)
Pulsed TIG (GTAW-P)
Stick (SMAW)
Air carbon arc (CAC-A)
400: 1/4 in. maximum
800: 3/8 in. maximum

Input Power 208–575 V, 3- or 1-phase power

Amperage Range **400:** 3–400 A
800: 5–800 A

Rated Output **400:** 300 A at 32 V, 60% duty cycle
800: 600 A at 44 V, 60% duty cycle

Net Weight **400:** 134 lb. (61 kg)
800: 198 lb. (90 kg)



Allows for any input voltage hookup (208–575 V) with no manual linking, providing convenience in any job setting. Ideal solution for dirty or unreliable power.

Meter calibration allows digital meters to be calibrated for certification.

Cooler Power Supply (CPS) is an integrated 120-volt dedicated-use receptacle for the Coolmate™ 3.5.

Cooler-On-Demand™ feature operates the auxiliary cooling system only when needed, reducing noise, energy use, and airborne contaminants pulled through the cooler.

Wind Tunnel Technology™ protects internal electrical components from airborne contaminants, extending the product life.

Fan-On-Demand™ power source cooling system operates only when needed, reducing noise, energy use and the amount of contaminants pulled through the machine.

New!



Maxstar 400 machine only

Maxstar 400 Wireless Complete



Lift-Arc™ provides DC arc initiation without the use of high frequency.

Blue Lightning™ high-frequency (HF) arc starter for non-contact arc initiation. Provides more consistent arc starts and greater reliability compared to traditional HF arc starters.

DC TIG Features

Exceptionally smooth and precise arc for welding exotic materials.

Pulse. Pulsing can increase puddle agitation, arc stability and travel speeds while reducing heat input and distortion. These models provide extended ranges.

DC Stick Features

DIG control allows the arc characteristics to be changed for specific applications and electrodes. Lower the DIG setting for smooth running electrodes like E7018 and increase the DIG setting for stiffer, more penetrating electrodes like E6010.

Hot Start™ adaptive control provides positive arc starts without sticking.

Program memory features nine independent program memories that maintain/save your parameters.

Auto-postflow adjusts the length of postflow time based on the amperage setting, shielding your tungsten and eliminating the need to set the postflow time.



Power source is warranted for three years, parts and labor.



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An ITW Welding Company
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MillerWelds.com
f t y i n



Specifications (Subject to change without notice.)



Model	Input Power	Welding Amperage Range	Rated Output	Amps Input at Rated Load Output, 50/60 Hz							Max. Open-Circuit Voltage	Dimensions	Net Weight
Maxstar 400	3-phase	3–400 A	250 A at 30 V, 100% duty cycle	27	24	14	12	9	9.4	9.1	75 VDC (10–15 VDC*)	H: 24.75 in. (629 mm) W: 13.75 in. (349 mm) D: 22 in. (559 mm) with TIGRunner® H: 43.125 in. (1,095 mm) W: 23.125 in. (587 mm) D: 43.75 in. (1,111 mm)	134 lb. (61 kg) with TIGRunner® 251 lb. (114 kg)
			300 A at 32 V, 60% duty cycle	33	30	17	15	12	12	11.6			
	1-phase	3–400 A	200 A at 28 V, 100% duty cycle	36	33	18	15	12	7.4	6.9			
			250 A at 30 V, 60% duty cycle	48	43	24	20	16	10	9.2			
Maxstar 800	3-phase	5–800 A	500 A at 40 V, 100% duty cycle	68	61	34	30	24	24	23	75 VDC (10–15 VDC*)	H: 34.5 in. (876 mm) W: 13.75 in. (349 mm) D: 22 in. (559 mm)	198 lb. (90 kg)
			600 A at 44 V, 60% duty cycle	90	80	45	39	31	32	31			
	1-phase	5–800 A	400 A at 36 V, 100% duty cycle	89	80	44	38	30	19	17			
			500 A at 40 V, 60% duty cycle	126	112	61	53	41	26	24			

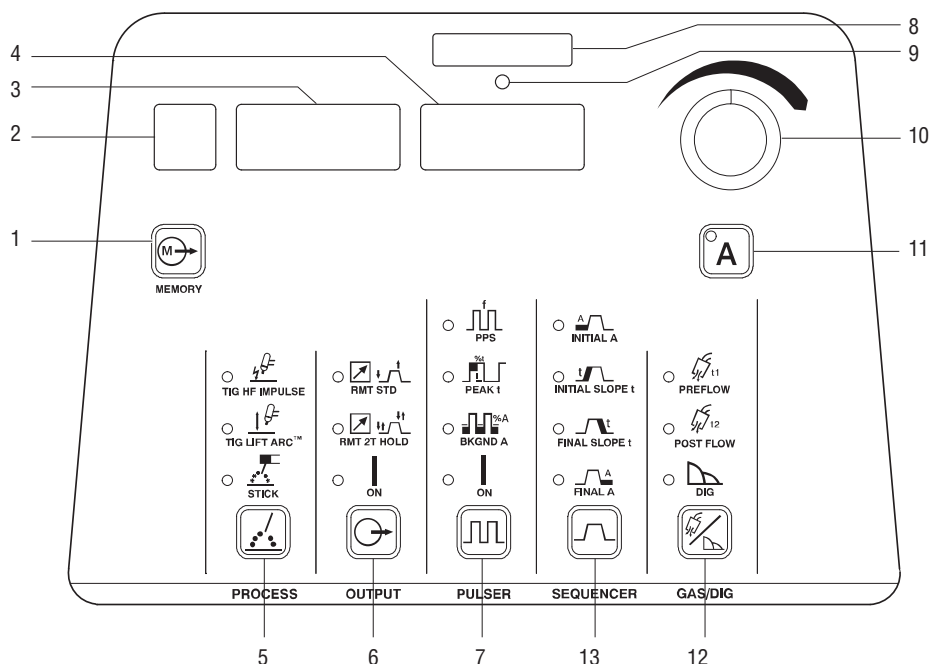
Certified by Canadian Standards Association to both the Canadian and U.S. Standards. All CE models conform to the applicable parts of the IEC 60974 series of standards.

*Indicates sense-voltage for Lift-Arc™ TIG and Low OCV stick.

Performance Data

Model	Input Power	TIG (GTAW) Duty Cycle	Stick (SMAW) Duty Cycle	DC TIG Material Thickness Range	Stick Electrode Maximum Diameter	Carbon Arc Gouging (CAC-A) Maximum	Generator Requirement
Maxstar 400	3-phase	400 A, 20% 300 A, 60% 250 A, 100%	400 A, 20% 300 A, 60% 250 A, 100%	.012–5/8 in. (0.3–15.9 mm)	6010: 1/4 in. (6.4 mm) 7018: 1/4 in. (6.4 mm) 7024: 1/4 in. (6.4 mm)	1/4 in. (6.4 mm)	20 kVA
	1-phase	300 A, 20% 250 A, 60% 200 A, 100%	300 A, 20% 250 A, 60% 200 A, 100%				
Maxstar 800	3-phase	800 A, 20% 600 A, 60% 500 A, 100%	800 A, 20% 600 A, 60% 500 A, 100%	.020–1 in. (0.5–25.4 mm)	6010: 1/4 in. (6.4 mm) 7018: 1/4 in. (6.4 mm) 7024: 1/4 in. (6.4 mm)	3/8 in. (9.5 mm)	45 kVA
	1-phase	500 A, 60% 400 A, 100%	500 A, 60% 400 A, 100%				

Maxstar® 400 and 800 Control Panel



Control Panel Parameter Values

1. Memory Switch	18 Combinations (9 DC TIG) (9 DC stick)	8. Memory Card Port	
2. Memory Display		9. Activity Indicator	
3. Voltmeter Display		10. Encoder Control	
4. Ammeter Display		11. Amperage Button	
5. Process/Arc Starting	TIG: HF impulse, Lift-Arc STICK: Adaptive Hot Start	12. Gas/DIG	
6. Output Control	Standard remote, 2T trigger hold, Output on	Preflow	Off–25.0 seconds
7. Pulser Control		Postflow	Auto/Off–50 seconds
Pulses per Second*	DC: 0.1–5,000 PPS	DIG*	Off–100%
Peak Time*	5–95%	13. Sequencer Control	
Background Amps*	5–95%	Initial Amps	3–400 A/5–800 A
		Initial Time	Off–25.0 seconds
		Initial Slope	Off–50.0 seconds
		Weld Time	Off–999 seconds
		Final Slope	Off–50.0 seconds
		Final Amps	3–400 A/5–800 A
		Final Time	Off–25.0 seconds

*Pro-Set parameter selectable.

User Menu (Press Gas and Amperage buttons.)

- Tungsten Size 400 = .020–3/16 in./GEN or 0.5–4.8 mm
800 = .040–1/4 in./GEN or 1.0–6.4 mm
- Remote Trigger = 2T/3T/4T/4TL/4TE/4Tm
- Stick Hot Start = ON/OFF

Tech Menu (Hold Gas and Amperage buttons five seconds.)

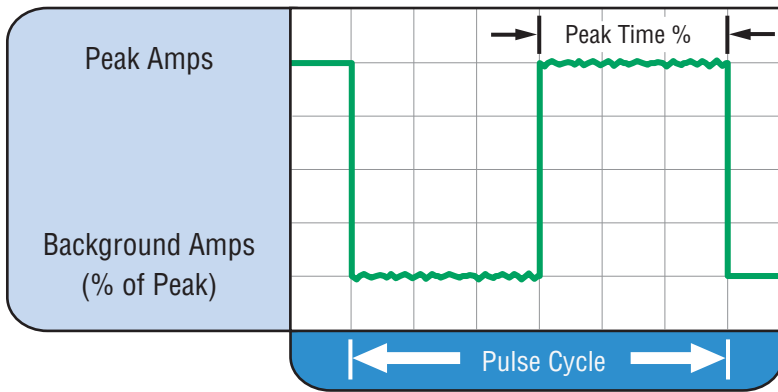
- Arc Time 0.0–9,999 hours
0.0–59 minutes
0–999,999 cycles
Resettable
- Error Log = Error event recorder
- Stick Stuc = OFF/ON
- OCV = LOW/NORM
- Weld Timers = OFF/ON
- Cooler Power = AUTO/ON/OFF
- Locks = OFF/1–4
- Meter Display
- External Pulse Control = OFF/ON
- Machine Reset
- Software Number
- Serial Number
- Slave (with Modbus® automation expansion) Address = 1–247
Baudrate = 9600/19.2K
Parity = EVEN/ODD/NONE

Pulsed TIG Controls

High-Speed DC Pulsed TIG Controls

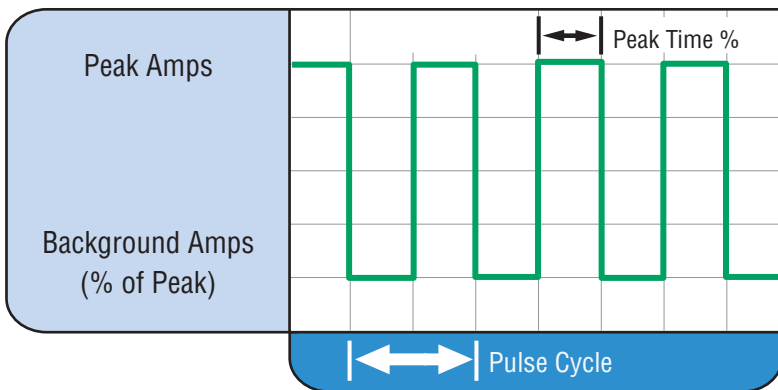
- **PPS Pulses per second (Hz):** DC = 0.1–5,000 PPS
- **% ON – % Peak Time:** 5–95% (Controls the amount of time during each pulse cycle at the PEAK amperage.)
- **Background Amps:** 5–99% (Sets the low-pulse amperage value as a % of the Peak Amps.)

Conventional Pulsed TIG



Typically from 1 to 10 PPS. Provides a heating and cooling effect on the weld puddle and can reduce distortion by lowering the average amperage. This heating and cooling effect also produces a distinct ripple pattern in the weld bead. The relationship between pulse frequency and travel speed determines the distance between the ripples. Slow pulsing can also be coordinated with filler metal addition and can increase overall control of the weld puddle.

High-Speed Pulsed TIG



In excess of 40 PPS, Pulsed TIG becomes more audible than visible — causing increased puddle agitation for a better as-welded microstructure.

Pulsing the weld current at high speeds — between a high Peak and a low Background amperage — can also constrict and focus the arc. This results in maximum arc stability, increased penetration and increased travel speeds (Common Range: 100–500 PPS).

The Arc-Sharpening effects of high speed pulsing are expanded to new dimensions. The ability to pulse at 5,000 PPS further enhances arc stability and concentration potential — which is extremely beneficial to automation where maximum travel speeds are required.

Maxstar® 400 and 800 Models/Packages

Machines and Preconfigured Water-Cooled Packages

Order machine only or use a single stock number to order a complete preconfigured system.



907716 and 907718 models shown.



907716001 package shown.



951693 package shown.

Machine Only		TIGRunner® Package (Machine/Cart/Cooler)		Complete Package (Machine/Cart/Cooler/Torch Kit/Remote)	
Maxstar 400 (CSA)	907716	Maxstar 400 (CSA)	907716001	Maxstar 400 (CSA) w/Foot Control	951692
Maxstar 400 (CE)	907716002			Maxstar 400 (CSA) w/Wireless Foot Control	951693
Maxstar 800 (CSA)	907718				
Maxstar 800 (CE)	907718002				
Comes with:		Comes with:		Comes with:	
• 8 ft. power cord (no plug) (400 model) • Two 50 mm Dinse-style connectors (400 model) • Two thread-lock connectors (800 model) • One thread-lock water-cooled connector (800 model)		• 8 ft. power cord (no plug) (400 model) • Runner™ cart 300244 • Coolmate™ 3.5 300245		• 8 ft. power cord (no plug) (400 model) • Runner™ cart 300244 • Coolmate™ 3.5 300245 • 4 gallons of coolant 043810 • W-375 torch kit 301268 • Remote control 194744 (foot) or - Remote control 300429 (wireless foot)	

Build a Water-Cooled Package

Select desired stock number for each step.



907716001 Maxstar 400 TIGRunner® shown with four bottles of 043810 Low-Conductivity Coolant.



194744 remote shown.



301268 kit shown.

Step #1 • Select Maxstar TIGRunner® and Coolant		Step #2 • Select Remote Control		Step #3 • Select Torch Kit	
Maxstar 400 TIGRunner	907716001	Wireless Foot	300429	W-250 Kit	300185
Maxstar 800	907718	RFCS-14 HD Foot	194744	W-280 Kit	300990
(add 300244 cart and 300245 Coolmate to create Maxstar 800 TIGRunner)		RCC-14 E/W Fingertip	151086	W-375 Kit	301268
		RCCS-14 N/S Fingertip	043688	(recommended for 400 model)	
		RMS-14 Pushbutton	187208	W-400 (WP-18SC) Kit	300186
		RMLS-14 Momentary/Maintained	129337	(recommended for 800 model)	
Low-Conductivity Coolant	043810	RHC-14 Hand	242211020		
(must be ordered in quantities of four)		Wireless Hand	300430		

Genuine Miller® Accessories

Water-Cooled Torch Kits



W-280 Torch Kit 300990

- Weldcraft™ W-280 25-foot (7.6 m) TIG torch with Dinse-style connector
- Torch cable cover
- Work clamp with 15-foot (4.6 m) 1/0 cable and Dinse-style connector
- Flowmeter regulator
- Gas hose (regulator to machine)
- AK4C torch accessory kit includes nozzles, collets, collet bodies and 2% ceriated tungsten electrodes (1/16, 3/32 and 1/8 inch)



W-375 Torch Kit 301268

Recommended for Maxstar 400

- Weldcraft™ W-375 25-foot (7.6 m) TIG torch with Dinse-style connector
- Torch cable cover
- Work clamp with 15-foot (4.6 m) 1/0 cable and Dinse-style connector
- Flowmeter regulator
- Gas hose (regulator to machine)
- AK4C torch accessory kit includes nozzles, collets, collet bodies and 2% ceriated tungsten electrodes (1/16, 3/32 and 1/8 inch)



W-400 (WP-18SC) Torch Kit 300186

Recommended for Maxstar 800

- Weldcraft™ W-400 (WP-18SC) 25-foot (7.6 m) TIG torch with thread-lock connector
- Torch cable cover
- Work clamp with 12-foot (3.7 m) 4/0 cable with thread-lock connector
- Flowmeter regulator
- Gas hose (regulator to machine)
- AK18C torch accessory kit includes nozzles, collets, collet bodies and 2% ceriated tungsten electrodes (3/32, 1/8 and 5/32 inch)



Water-Cooled TIG Torch Connector 195377

For Dynasty® and Maxstar® 400.

50 mm Dinse-style with water return line. For use with all Weldcraft™ water-cooled torches.



Water-Cooled TIG Torch Connector 225028

For Dynasty and Maxstar 800.

50 mm thread-lock with water return line. For use with all Weldcraft™ water-cooled torches.



Runner™ Cart 300244

Designed to accommodate Dynasty or Maxstar 400 or 800 power sources and a Coolmate™ 3.5 Cooler. Cart features single cylinder rack, foot pedal holder, three cable/torch holders, and two TIG electrode filler holders.



Coolmate™ 3.5 300245

Designed to integrate with the Dynasty and Maxstar 400 and 800 power sources. For use with water-cooled torches rated up to 600 amps. 3.5-gallon capacity.



Low-Conductivity TIG Coolant 043810

Sold in multiples of four in one-gallon recyclable plastic bottles. Miller coolants contains a base of ethylene

glycol and deionized water to protect against freezing to -37° Fahrenheit (-38°C) or boiling to 227° Fahrenheit (108°C).

Automation Interface Connection Kit 278161 Field

Provides control of power source welding parameters through a 28-pin receptacle. The 28-pin receptacle replaces the standard 14-pin receptacle and requires a PLC controller to operate the power source. Ideal for automated equipment integration.

Weld Current Sensor 300179 Field

Detects when work clamp is not connected and prevents expensive damage to disconnect devices and input power cord and wiring.

Genuine Miller® Accessories (Continued)

Remote Controls and Switches



Wireless Remote Foot Control 300429

For remote current and contactor control. Receiver plugs directly into the 14-pin

receptacle of Miller machine. 90-foot (27.4 m) operating range.



Wireless Remote Hand Control 300430

For remote current and contactor control. Receiver plugs directly into the 14-pin receptacle of Miller machine. 300-foot (91.4 m) operating range.



RCCS-14 Remote Contactor and Current Control 043688

North/south rotary-motion fingertip control attaches to TIG torch using two hook-and-loop fasteners. Great for applications that require a finer amperage control. Includes 26.5-foot (8 m) cord and 14-pin plug.



RCC-14 Remote Contactor and Current Control 151086

East/west rotary-motion fingertip control attaches to TIG torch using two hook-and-loop fasteners. Great for production or contractors that need quick ramp-up. Includes 26.5-foot (8 m) cord and 14-pin plug.



RFCS-14 HD Foot Control 194744

Maximum flexibility is accomplished with a reconfigurable cord that can exit the front, back or either side of

the pedal. Foot pedal provides remote current and contactor control. Includes 20-foot (6 m) cord and 14-pin plug.



RHC-14 Hand Control 242211020

Miniature hand control for remote current and contactor control. Dimensions: 4 x 4 x 3.25 inches (102 x 102 x 83 mm).

Includes 20-foot (6 m) cord and 14-pin plug.



RMLS-14 Switch 129337

Momentary- and maintained-contact rocker switch for contactor control. Push forward for maintained contact and backward for momentary contact. Includes 26.5-foot (8 m) cord and 14-pin plug.



RMS-14 On/Off Control 187208

Momentary-contact switch for contactor control. Rubber-covered pushbutton dome switch ideal for repetitive on-off applications. Includes 26.5-foot (8 m) cord and 14-pin plug.

Extension Cables for 14-Pin Remote Controls

242208025 25 ft. (7.6 m)

242208050 50 ft. (15.2 m)

242208080 80 ft. (24.4 m)



Performance TIG Gloves

263345 X-Small

263346 Small

263347 Medium

263348 Large

263349 X-Large

Completely unlined, goat grain leather with triple-padded palm.



Performance TIG/Multitask Gloves

263352 Small

263353 Medium

263354 Large

263355 X-Large

Goat grain leather with dual-padded palm and wool back.

Memory Cards

Memory Card Expansion

301151 14-pin automation expansion

Provides the ability to access common automation functions through the 14-pin connection.

301152 14-pin Modbus® expansion

Provides the ability to access basic and advanced functions through the 14-pin connection.

Memory Card (Blank) 301080

A blank, commercially available memory card used for transferring software updates and expandable features from your computer to the machine.

Free software updates and feature expansions can be downloaded at MillerWelds.com/tigsoftware.

Educational Materials

To order, please call Miller Literature at 866-931-9732 or visit MillerWelds.com/resources/tools.

Gas Tungsten Arc Welding (TIG) Publication 250833

Tungsten

Tungsten	Amp Range	2% Ceriated (AC/DC)	2% Lanthanated (AC/DC)
1/16 in. (1.6 mm)	70–150 A	WC116X7	WL2116X7
3/32 in. (2.4 mm)	140–250 A	WC332X7	WL2332X7
1/8 in. (3.2 mm)	225–400 A	WC018X7	WL2018X7
5/32 in. (4.0 mm)	300–500 A	WC532X7	WL2532X7

Ordering Information

Equipment and Options	Stock No.	Description	Qty.	Price
Maxstar® 400	907716	Auto-Line™ 208–575 V, 50/60 Hz, CSA . 8 ft. power cord		
Maxstar® 400 International	907716002	Auto-Line™ 380–575 V, 50/60 Hz, CE . 8 ft. power cord		
Maxstar® 400 TIGRunner®	907716001	Auto-Line™ 208–575 V, 50/60 Hz, CSA . 8 ft. power cord. <i>Requires coolant</i>		
Maxstar® 400 Wireless Complete	951693	Auto-Line™ 208–575 V, 50/60 Hz, CSA . 8 ft. power cord		
Maxstar® 400 Complete	951692	Auto-Line™ 208–575 V, 50/60 Hz, CSA . 8 ft. power cord		
Maxstar® 800	907718	Auto-Line™ 208–575 V, 50/60 Hz, CSA		
Maxstar® 800 International	907718002	Auto-Line™ 380–575 V, 50/60 Hz, CE		
TIG Torches, Kits and Connectors				
Water-Cooled Torch Kits	300990 301268 300186	W-280 W-375 (recommended for Maxstar 400) W-400 (WP-18SC) (recommended for Maxstar 800)		
Water-Cooled TIG Torch Connectors	195377 225028	Connects Weldcraft™ water-cooled torches to Dinse-style connector Connects Weldcraft™ water-cooled torches to Maxstar 800 (thread-lock connector included with 800 models)		
Tungsten		See page 7		
Remote Controls				
Wireless Remote Foot Control	300429	Foot control with wireless 90 ft. (27.4 m) operating range		
Wireless Remote Hand Control	300430	Hand control with wireless 300 ft. (91.4 m) operating range		
RCCS-14	043688	North/south fingertip control		
RCC-14	151086	East/west fingertip control		
RFCS-14 HD	194744	Heavy-duty foot control		
RHC-14	242211020	Hand control		
RMLS-14	129337	Momentary/maintained rocker switch		
RMS-14	187208	Momentary rubber dome switch		
Extension Cables		See page 7		
Accessories				
Runner™ Cart	300244	See page 6		
Coolmate™ 3.5	300245	120 V, 50/60 Hz, CE . <i>Requires coolant</i>		
TIG Coolant (Must be ordered in quantities of four)	043810	1-gallon plastic bottle. Protects against freezing to -37° Fahrenheit (-38°C) or boiling to 227° Fahrenheit (108°C)		
Automation Interface Kit	278161	Field installation required. Provides 28-pin automation connections		
Weld Current Sensor	300179	Field installation required. Detects when work clamp is not connected		
Dinse-Style Connector 50 mm (1 male)	042418	Used to connect weld cable to Dinse terminal machine		
Thread-Lock Connectors (2 male)	225029	Used to connect weld cable to Dynasty 800 or Maxstar 800		
Dinse-Style Connector 50 mm (1 male, 1 female)	042419	Used to extend weld cables		
Dinse/Tweco® Adapter	042465	Male Dinse to female Tweco		
Dinse/Cam-Lok Adapter	042466	Male Dinse to female Cam-Lok		
TIG Welding Gloves		See page 7		
Memory Cards	301151 301152 301080	14-Pin automation expansion 14-Pin Modbus® expansion Blank		
Gas Tungsten Arc Welding (TIG) Publication	250833			

Date:

Total Quoted Price:

Distributed by:

